

《大学物理 II》期末考试卷 (B) 答案

一、单选题 〔每个题 2 分，共计 30 分〕

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
C	B	A	D	A	D	D	C	D	D	D	A	B	B	A

二、填空题 〔每空 3 分，共计 30 分〕

1. 1

2. $\frac{g}{L}$

3. $\frac{\lambda_1 + \lambda_2}{2\pi\epsilon_0 r}$

4. $-B\pi r^2 \cos \alpha$

5. $-I$

6. $\frac{4}{3}$

7. 660

8. 亮条纹

9. 7

10. $\frac{hc}{\lambda_0} - \frac{hc}{\lambda}$

三、计算题 〔每题各 10 分，共计 40 分〕

1.

$$(T_2 - T_1)r = J_2\beta_2$$

$$T_1R = J_1\beta_1$$

$$mg - T_2 = ma$$

$$a = r\beta_2$$

$$a = R\beta_1$$

$$a = \frac{mg}{\frac{1}{2}(M_1 + M_2) + m} = 4m/s^2$$

$$T_1 = \frac{1}{2}M_1a = 48N$$

$$T_2 = mg - ma = 60N$$

2. 解: (1) A→B:

$$A_1 = \frac{1}{2}(p_B + p_A)(V_B - V_A) = 200 J$$

$$\Delta E_1 = \nu C_v(T_B - T_A) = 3(p_B V_B - p_A V_A) / 2 = 750 J$$

$$Q_1 = A_1 + \Delta E_1 = 950 J$$

B→C:

$$A_2 = 0$$

$$\Delta E_2 = \nu C_v(T_C - T_B) = 3(p_C V_C - p_B V_B) / 2 = -600 J$$

$$Q_2 = A_2 + \Delta E_2 = -600 J$$

C→A:

$$A_3 = p_A(V_A - V_C) = -100 J$$

$$\Delta E_3 = \nu C_v(T_A - T_C) = \frac{3}{2}(p_A V_A - p_C V_C) = -150 J$$

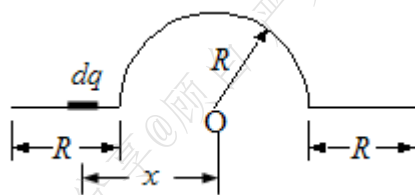
$$Q_3 = A_3 + \Delta E_3 = -250 J$$

(2)

$$A = A_1 + A_2 + A_3 = 100 J$$

$$Q = Q_1 + Q_2 + Q_3 = 100 J$$

3. 解



$$U_1 = \int \frac{dq}{4\pi\epsilon_0 R} = \int_0^\pi \frac{\lambda d\theta}{4\pi\epsilon_0} = \frac{\lambda}{4\epsilon_0}$$

$$U_2 = 2 \int \frac{dq}{4\pi\epsilon_0 x} = 2 \int_R^{2R} \frac{\lambda dx}{4\pi\epsilon_0 x} = \frac{\lambda}{2\pi\epsilon_0} \ln 2$$

$$U = U_1 + U_2 = \frac{\lambda}{2\pi\epsilon_0} \ln 2 + \frac{\lambda}{4\epsilon_0}$$

4. 解: (1) $\Phi(t) = \iint_s \vec{B} \cdot d\vec{s} = \int \frac{\mu_0 I}{2\pi r} l dr = \frac{\mu_0 I l}{2\pi} \int_{a+vt}^{b+vt} \frac{dr}{r} = \frac{\mu_0 I l}{2\pi} \ln \frac{b+vt}{a+vt}$

(2) 依据法拉第电磁感应定律得 $\varepsilon = -\frac{d\Phi}{dt} = \frac{\mu_0 I l}{2\pi} \cdot \frac{(a-b)v}{(b+vt)(a+vt)} \Big|$

$$\varepsilon = -\frac{d\Phi}{dt} \Big|_{t=0} = \frac{\mu_0 I l v (b-a)}{2\pi ab}$$